AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

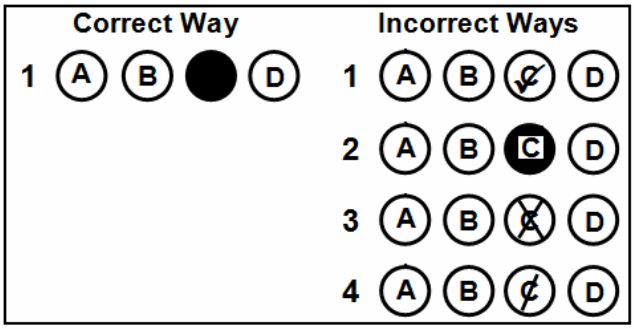
MAY 2015

Physics Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.

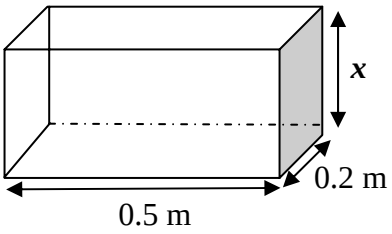


Candidate's Signature

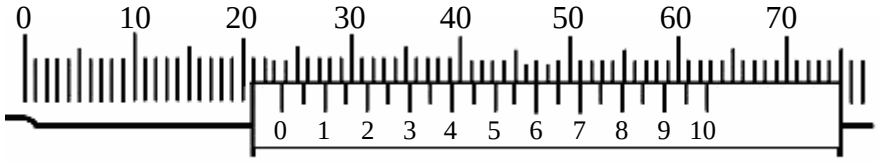
5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. If the volume of the given cuboid is 0.03 m^3, then the value of ***x*** will be

A. 0.1 m
B. 0.2 m
C. 0.3 m
D. 0.5 m



2. In the given diagram showing part of a Vernier calliper the Vernier scale reading is

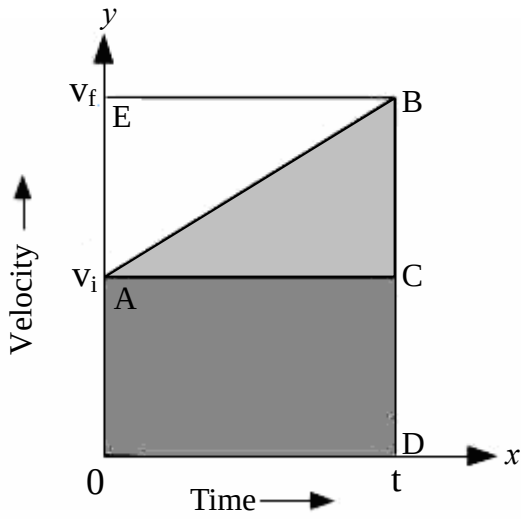


A. 0.06 mm
B. 0.6 mm
C. 6 mm
D. 60 mm

3. Which of the following is the most accurate instrument for measuring length?

A. Meter rule
B. Screw gauge
C. Measuring tape
D. Vernier calliper

4. The below given picture shows the velocity-time graph for the motion of a body.



The slope of the line segment AB in the above graph shows the

A. acceleration.
B. displacement of the body.
C. total distance covered by the body.
D. total time taken for the motion of the body.

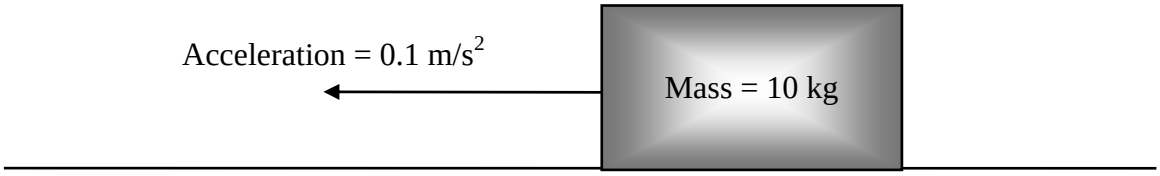
5. Which of the following is NOT a vector quantity?

   A. Power
   B. Torque
   C. Tension
   D. Momentum

6. The distance covered in unit time is known as

   A. speed.
   B. velocity.
   C. gravitation.
   D. acceleration.

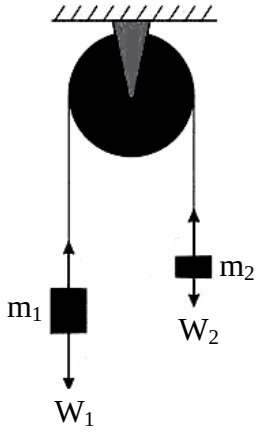
7. The value of the acting force in the given diagram is



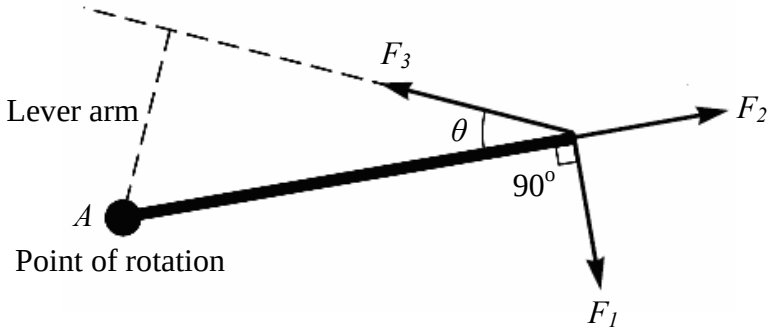
   A. 1 N
   B. 10 N
   C. 100 N
   D. 1000 N

8. If $W_1 > W_2$ in the given pulley then which of the following options must be TRUE for the direction of the acceleration for m_1 and m_2?

	Acceleration of m_1	**Acceleration of m_2**
A	downward	downward
B	upward	upward
C	downward	upward
D	upward	downward

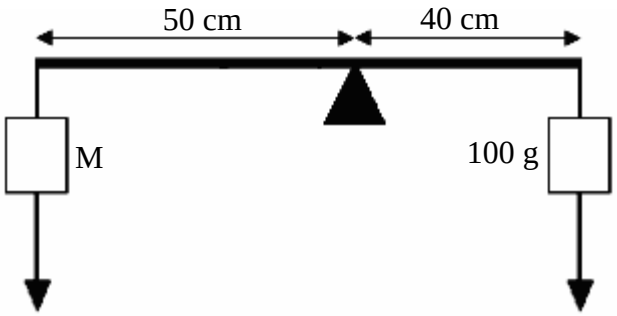


9. If all three forces F_1, F_2 and F_3 are equal in magnitude then which of the following force/s would individually produce zero torque about A?



A. F_1 only
B. F_2 only
C. F_3 only
D. F_2 and F_3

10. A meter rule is pivoted as shown in the given figure. In order to balance the meter rule, the value of mass M should be



A. 80 g
B. 100 g
C. 150 g
D. 220 g

11. The rotational effect of forces is called

A. velocity.
B. moment.
C. momentum.
D. acceleration.

12. The weight of an object on the Earth is the result of the gravitational field of attraction between the

A. Sun and the object.
B. Earth and the Moon.
C. Earth and the object.
D. Moon and the object.

13. Which of the following correctly describes the variation of gravity (g) with altitude?

    A. It is directly proportional to the distance from the centre of earth.
    B. It is inversely proportional to the distance from the centre of earth.
    C. It is directly proportional to the square of the distance from the centre of earth.
    D. It is inversely proportional to the square of the distance from the centre of earth.

14. Planets revolve around the Sun due to a

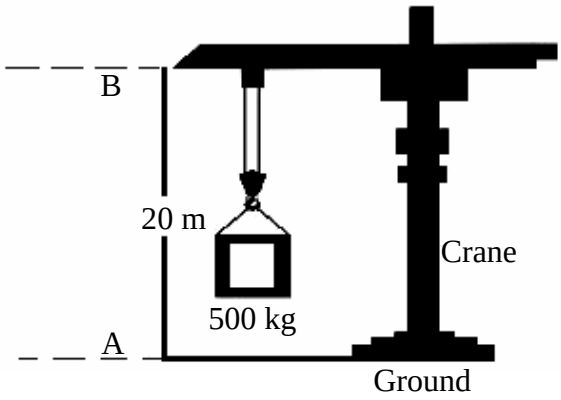
    A. frictional force.
    B. cohesive force.
    C. electrostatic force.
    D. gravitational force.

15. When a spring is compressed, it possesses

    A. heat energy.
    B. sound energy.
    C. kinetic energy.
    D. elastic potential energy.

16. In the given figure a crane is lifting a load of 500 kg from the position A to B and the value of acceleration due to gravity is 9.8 m/s^2. The work done by the crane is

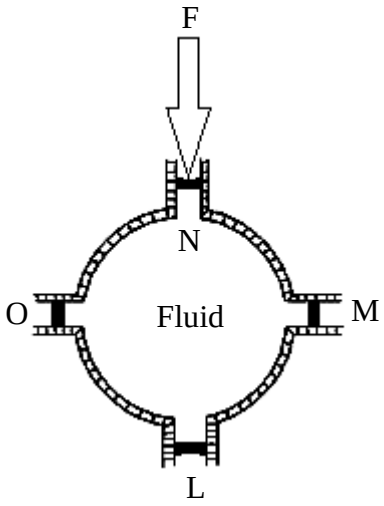
    A. 2.5 J
    B. 245 J
    C. 1020 J
    D. 98000 J



17. The energy possessed by a body by virtue of its motion is called

    A. heat energy.
    B. kinetic energy.
    C. potential energy.
    D. electrical energy.

18. A fluid vessel having equal sized holes at L, M, N and O are shown in the given diagram.



If a piston is pushed with a force F at N then the increase in pressure transmitted by the fluid will be

A. greatest at L.
B. greatest at M.
C. greatest at O.
D. same at L, M and O.

19. Which of the following is used to determine the density of an object?

A. Hooke's law
B. Pascal's law
C. Archimedes principle
D. Principle of moments

20. All of the following statements are the features of the kinetic molecular theory of matter EXCEPT

A. molecules attract each other.
B. matter is made up of molecules.
C. molecules remain in continuous motion.
D. pressure becomes zero when molecules collide with each other.

21. A thermometric liquid should have the following properties.

	Freezing Point	**Boiling Point**
A	low	high
B	low	low
C	high	high
D	high	low

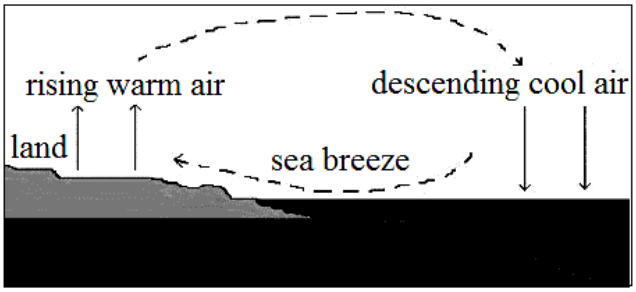
22. Which of the following is the only similarity between evaporation and boiling?

A. Both processes take place within the solid.
B. In both processes temperature remains constant.
C. In both processes bubbles are formed in the liquid.
D. Both processes involve change of state of liquid into vapour.

23. The transfer of energy from a hot object to a cold object is called

A. heat.
B. evaporation.
C. temperature.
D. internal energy.

24. The flow of cool air from the sea towards the land produces a sea breeze as shown in the below diagram.



The sea and land breezes are the result of

A. radiation.
B. convection.
C. conduction.
D. evaporation.

25. Wearing a white uniform keeps a student cool when the weather is hot. It is because the white colour is

A. dull and dark.
B. shiny and bright.
C. a good absorber of heat.
D. a good reflector of radiation.

END OF PAPER

Please use this page for rough work